

Sequential Hypothesis Testing for Black-Box Fairness Audits

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Audits of deployed AI systems often have to decide whether a system violates a behavioral requirement before the auditor can assemble a large, fixed test set. This paper develops a sequential testing framework for black-box fairness auditing. We formalize an audit as a pre-specified hypothesis test on a disparity functional and instantiate the procedure with Wald’s Sequential Probability Ratio Test (SPRT). The framework separates three choices that are often conflated in fairness evaluations: the fairness construct to be audited, the tolerance threshold that makes a disparity practically or legally material, and the statistical risks that the auditor is willing to incur. We evaluate the approach in two studies. In controlled simulations, sequential audits detect violations of statistical parity, equal opportunity, and equalized odds while typically using fewer samples than fixed-sample two-proportion tests under the studied designs. In a German Credit case study, the same protocol detects persistent disparities in a baseline classifier and in a classifier trained without the sensitive attribute, while producing substantially fewer bias flags after demographic-parity constrained training. The results show that sequential tests can make fairness audits more sample-efficient, but also that nominal error guarantees require careful pre-registration, treatment of composite hypotheses, multiple-testing correction, and validation under the actual sampling protocol. We further stress-test the procedure on a fair equalized-odds model and provide reporting recommendations for finite-budget audits.

Keywords: algorithmic fairness, AI auditing, sequential hypothesis testing, black-box audit, statistical parity, equalized odds

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1 Introduction

AI systems increasingly mediate access to credit, employment, public services, education, healthcare, and other consequential opportunities. This has made algorithmic auditing a central mechanism for identifying discriminatory behavior and producing evidence of compliance with organizational, legal, or regulatory expectations [1, 8, 14]. Yet many audits are conducted under severe information constraints. Auditors may have only black-box access to a deployed model, may be limited to a finite query or data-collection budget, and may have to audit several groups, time periods, or fairness criteria simultaneously. These constraints make purely fixed-sample evaluations inefficient and can encourage post hoc metric selection, repeated testing, and other practices that inflate false-positive rates.

The access restriction is not merely technical. It determines what kind of evidence the audit can produce. A black-box audit can document discriminatory input-output behavior, but it usually cannot identify whether the cause lies in the training data, optimization objective, deployment environment, or organizational process. For this reason, our method is designed as a risk-proportional screening tool: it can support monitoring and escalation decisions, but it should be embedded in a broader audit process when the system operates in a high-impact domain.

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This paper studies fairness auditing as a sequential statistical decision problem. Rather than requiring the auditor to fix the sample size in advance, a sequential audit updates evidence as observations arrive and stops once the evidence is sufficiently strong to reach a pre-specified decision. Wald’s Sequential Probability Ratio Test (SPRT) is a natural starting point: for two simple hypotheses and prescribed Type I and Type II error rates, it minimizes expected sample size among tests with the same error guarantees [20]. These properties are particularly attractive for audits where additional samples are costly, legally sensitive, or practically unavailable. Our central claim is not that sequential testing resolves the normative difficulty of defining fairness. Instead, it makes the statistical part of the audit explicit. A valid audit must specify the fairness metric, the protected group comparison, the materiality threshold, the data-collection protocol, the stopping rule, and the error rates before looking at the data. The threshold is especially important: a zero-disparity null is often unrealistic, while a tolerance threshold can encode a minimum disparity that is considered practically, legally, or ethically material. In this sense, the proposed framework is meant to support accountable audit design rather than to replace domain-specific judgment.

This paper makes three contributions. First, we formalize a sequential black-box audit protocol for group fairness metrics, including statistical parity, equal opportunity, and equalized odds. Second, we give an explicit likelihood-ratio construction for two-proportion fairness audits and discuss how it should be adapted to composite hypotheses, batched sampling, finite audit budgets, and multiple metrics. Third, we empirically evaluate the framework in controlled simulations and in a real-world credit-scoring case study. The expanded experiments quantify sample savings relative to fixed-sample references, examine the effect of the materiality threshold, and include a fair-model stress test that exposes when equalized-odds audits can deviate from nominal error rates. The results identify failure modes that are important for ECAF-style fairness research: metric choice matters, removing a sensitive feature is not sufficient to remove disparities, finite budgets create genuine inconclusive outcomes, and uncalibrated batched or multi-metric sequential testing can misstate statistical risk.

2 Related Work

Fairness criteria and their limits. Group fairness criteria compare model behavior across groups defined by protected or sensitive attributes. Statistical parity, equal opportunity, equalized odds, and predictive parity capture different normative and technical desiderata [1, 10, 15]. Because these criteria can conflict, metric selection is part of the audit question rather than a neutral implementation detail, and should be justified by the decision context, affected population, and relevant legal or institutional standard.

Algorithmic auditing under constrained access. Audit frameworks emphasize end-to-end accountability, documentation, and evaluation of deployed systems [16, 17]. Recent work cautions that black-box access is often insufficient to explain model behavior, especially when harms depend on data pipelines, deployment context, or organizational decisions [5, 6]. This limitation is amplified by model multiplicity: the same training data or development procedure can yield models with different fairness behavior [4]. We therefore treat black-box auditing as a statistically principled screening layer: a violation flag provides behavioral evidence and can trigger deeper investigation, while a no-flag outcome is not proof of fairness in all relevant senses.

Statistical testing for fairness. Several strands of work connect fairness evaluation to hypothesis testing, uncertainty quantification, and finite-sample guarantees [9, 19]. The statistical motivation also relates to anytime-valid inference, which uses likelihood-ratio and supermartingale arguments to support continuous monitoring without

invalidating error control [12, 13]. Many operational audits, however, still rely on fixed-sample estimates or confidence intervals on held-out data. Sequential methods provide a complementary design: the auditor commits to a stopping rule before sampling, controls risk at the design stage, and can stop early when evidence is clear, which is useful when fairness must be evaluated across subgroups, attributes, or time windows.

Recent work by Chugg et al. connects fairness assessment to sequential and anytime-valid testing through a testing-by-betting approach for deployed classification and regression models [7]. Our contribution is complementary: we focus on a black-box audit protocol based on pre-registered likelihood-ratio contrasts, materiality thresholds, finite audit budgets, multiplicity across fairness metrics, and reporting guidance for inconclusive outcomes.

Unlike fixed-sample tests or post hoc confidence-interval reporting, the proposed audit treats the stopping rule as part of the statistical design and gives the auditor an explicit finite-budget decision protocol. Unlike general sequential-testing results, our framework identifies the audit-specific choices, i.e., metric selection, materiality threshold, access model, multiplicity correction, and inconclusive outcomes, which determine whether a sequential result can support an accountability claim.

3 Audit Target and Fairness Metrics

3.1 Audit setting and access model

The audit is conducted after a system has been trained and deployed, with the auditor able to query or observe the final system. This choice is important because different access levels support different claims. Table 1 summarizes the access model used in the paper and clarifies what our sequential procedure does and does not cover.

This paper therefore assumes black-box behavioral access, not because it is sufficient for every accountability question, but because it is a common minimum-access setting. The resulting audit claim is intentionally narrow: for a specified population, metric, threshold, sampling design, and error budget, the auditor obtains evidence about whether the observed behavior violates the pre-specified fairness criterion.

3.2 Fairness metrics as auditable estimands

We consider a binary classification setting. Let $X \in \mathcal{X}$ denote features, $A \in \{a, b\}$ a binary sensitive attribute, $Y \in \{0, 1\}$ the outcome label, and $\hat{Y} = f(X) \in \{0, 1\}$ the prediction produced by a deployed classifier. In a black-box audit, the auditor observes inputs, outputs, and any labels required by the metric, but does not require access to model parameters or training data.

An audit target is a scalar or vector-valued disparity functional $D(f)$ computed from the joint distribution of $(X, A, Y, \hat{Y})$. The following examples illustrate the metrics used in our experiments:

$$\text{SPD}(f) = \mathbb{P}(\hat{Y} = 1 \mid A = a) - \mathbb{P}(\hat{Y} = 1 \mid A = b), \quad (1)$$

$$\text{EOP}(f) = \text{TPR}_a - \text{TPR}_b, \quad \text{TPR}_g = \mathbb{P}(\hat{Y} = 1 \mid A = g, Y = 1), \quad (2)$$

$$\text{EO}(f) = (\text{TPR}_a - \text{TPR}_b, \text{FPR}_a - \text{FPR}_b), \quad \text{FPR}_g = \mathbb{P}(\hat{Y} = 1 \mid A = g, Y = 0). \quad (3)$$

Statistical parity can be audited without outcome labels, whereas equal opportunity and equalized odds require labels for the relevant conditional populations. When $D(f)$ is vector-valued, a violation means that at least one

Table 1. Access levels and their relationship to the proposed sequential audit. The method in this paper uses the last row: behavioral access to the deployed system.

Access level	Evidence provided	Relation to the proposed audit
Training data	Representation, label quality, sampling bias, and potential proxy variables	Useful for diagnosis, but does not determine the final deployed behavior.
Training procedure	Objectives, constraints, model class, and optimization choices	Helps assess developer decisions, but cannot by itself certify realized behavior.
Model architecture or internals	Structural design, parameters, or explanations of model behavior	Supports deeper causal or mechanistic analysis, but may be unavailable to independent auditors.
Black-box deployed model	Input-output behavior for specified populations and conditions	The setting addressed here: sequential tests provide statistically disciplined behavioral evidence.

component exceeds its tolerance threshold. Equalized odds is vector-valued, so it is implemented as a family of tests, one for the TPR contrast and one for the FPR contrast.

When the disadvantaged direction is not known in advance, a two-sided audit can be implemented as two pre-specified one-sided sequential tests, one for each direction of disparity, with the Type I error allocated across directions. This avoids choosing the disadvantaged group after observing the audit data. In the empirical sections below, the metric, protected-group contrast, and anticipated direction of disparity encoded by the design point are fixed before sampling; for absolute-disparity claims, we interpret the reported procedure as a pre-specified two-direction design with the stated multiplicity caveats.

We write the audit question as

$$H_0 : |D(f)| \leq \delta_0 \quad \text{versus} \quad H_1 : |D(f)| \geq \delta_1, \quad (4)$$

where $0 \leq \delta_0 < \delta_1$. The interval (δ_0, δ_1) is an indifference region: the audit is not designed to make a high-confidence decision for disparities in this band. In the experiments below, we use the common design choice $\delta_0 = 0$ and $\delta_1 = \delta$, while emphasizing that a deployed audit should justify δ before sampling. A smaller δ makes the audit more sensitive but typically increases sample requirements; a larger δ reduces sample requirements but may ignore practically important disparities. The separation $\delta_0 < \delta_1$ is intentional. Setting $\delta_0 = \delta_1$ would remove the indifference region and would require the audit to distinguish distributions arbitrarily close to a single boundary, which can lead to long stopping times and unstable finite-sample operating characteristics. If the true disparity lies in (δ_0, δ_1) , neither hypothesis is the intended data-generating description; the test may stop at either boundary, and the result should be interpreted as a decision relative to the pre-specified design rather than as a definitive classification of fairness.

4 Sequential Fairness Audit

4.1 Two-proportion likelihood ratio

Many group fairness metrics reduce to comparing two Bernoulli probabilities. Let $Z_{g,i} \in \{0, 1\}$ be the Bernoulli event relevant for group $g \in \{a, b\}$: a positive prediction for statistical parity, a true positive among $Y = 1$ cases for equal opportunity, or a false positive among $Y = 0$ cases for one component of equalized odds. Denote the corresponding probabilities by p_a and p_b .

For a one-sided audit of whether group a receives a larger event probability than group b , a design point can be specified as

$$H_0 : (p_a, p_b) = (p_0, p_0) \quad \text{and} \quad H_1 : (p_a, p_b) = (p_0 + \delta/2, p_0 - \delta/2), \quad (5)$$

with probabilities truncated away from 0 and 1 when necessary. In practice, p_0 must be fixed before the sequential audit because the operating characteristics of the likelihood ratio depend on this design value. In the synthetic experiments, p_0 is set from the design model as the pooled event probability for the relevant Bernoulli contrast. In the German Credit experiments, p_0 is computed before the audit from the trained model's pooled positive-prediction rate on the training split; it is then held fixed throughout all sequential audit runs. No observation from the sequential audit set is used to estimate or update p_0 . The design points should not be interpreted as estimates of the unknown true probabilities. They are operating points used to construct the likelihood-ratio increments and therefore determine the drift, power, and expected stopping time of the sequential test. If the true probabilities are close to the null design point, the log-likelihood ratio tends to drift downward; if they are close to the alternative design point, it tends to drift upward. When the true distribution differs from both design points, the SPRT remains a pre-specified sequential decision rule, but its nominal operating characteristics should be evaluated by calibration under the intended sampling protocol. A two-sided audit can run two one-sided tests with adjusted error rates, or can use a mixture or generalized likelihood-ratio construction. Our implementation follows the transparent one-sided building block: the direction of concern, the metric, the design value p_0 , and the design gap δ are fixed before the audit.

After observing n_a relevant observations from group a and n_b from group b , with s_a and s_b successes, the log-likelihood ratio for the simple design points is

$$S_n = s_a \log \frac{p_{1a}}{p_{0a}} + (n_a - s_a) \log \frac{1 - p_{1a}}{1 - p_{0a}} \quad (6)$$

$$+ s_b \log \frac{p_{1b}}{p_{0b}} + (n_b - s_b) \log \frac{1 - p_{1b}}{1 - p_{0b}}, \quad (7)$$

where (p_{0a}, p_{0b}) and (p_{1a}, p_{1b}) are the null and alternative design probabilities. The statistic updates additively whenever a relevant observation is collected.

4.2 Stopping rule and error rates

Given Type I error α and Type II error β , the SPRT uses thresholds

$$A = \frac{1 - \beta}{\alpha}, \quad B = \frac{\beta}{1 - \alpha}. \quad (8)$$

The audit flags a violation when $S_n \geq \log A$, terminates at the lower boundary without flagging a violation when $S_n \leq \log B$, and otherwise continues sampling. For example, $\alpha = 0.05$ and $\beta = 0.20$ yield $A = 16$ and $B \approx 0.2105$.

When the audit uses batches rather than single observations, the same rule is applied after each batch, but overshoot can make the realized error rates differ from the nominal values. This is one reason to report calibration results under the actual sampling procedure.

4.3 Multiple testing and audit validity

When an audit examines several groups, metrics, thresholds, or time windows, the family of tests must be specified in advance. To control family-wise error, the audit can allocate α/m to each of m tests using a Bonferroni correction or use a less conservative step-down correction such as Holm's procedure. If the audit objective tolerates a controlled fraction of false discoveries, false-discovery-rate procedures are also possible [2, 3]. Equalized odds is a simple example: because both TPR and FPR contrasts must satisfy the criterion, the audit should control the error rate across both component tests. If the two components were each tested at level $\alpha = 0.05$ without allocation, the family-wise false-positive probability could approach $1 - (1 - 0.05)^2 \approx 9.75\%$ under approximate independence, which is close to the false-flag rate observed in our fair equalized-odds stress test.

Family-wise control for multi-metric audits. If an audit consists of m pre-specified sequential tests and test j has Type I error at most α_j under its corresponding null, then the probability of at least one false violation flag is bounded by

$$\mathbb{P}_0\{\text{at least one false flag}\} \leq \sum_{j=1}^m \alpha_j. \quad (9)$$

Thus, choosing $\alpha_j = \alpha/m$ gives family-wise error control at level α by the union bound, regardless of dependence among tests. For a two-sided equalized-odds audit, this allocation must account for both the TPR and FPR components and for both directions of disparity if the disadvantaged direction is not fixed before sampling.

The validity of the sequential audit depends on the data-generating process and the sampling protocol. The strongest interpretation requires observations to be independent, identically distributed, and stationary over the audit period. Violations can arise from repeated observations of the same person, temporal drift, strategic adaptation, or audit-aware system behavior. These are not merely technical details: they determine whether the audit evidence can support a compliance claim.

4.4 A practical audit protocol

We use the following protocol in the empirical studies:

- (1) Pre-specify the fairness metric, protected groups, direction(s) of concern, materiality threshold δ , design value p_0 , sampling budget, batch size, and risk levels α, β .
- (2) Translate the metric into one or more Bernoulli contrasts and construct the corresponding likelihood-ratio updates.
- (3) Apply multiplicity correction when more than one contrast is tested.
- (4) Calibrate the planned protocol, including batch size and finite-budget handling, under plausible null and alternative scenarios before deployment.
- (5) Sample observations sequentially, update the log-likelihood ratio, and stop only when a pre-specified boundary is crossed or the audit budget is exhausted.

- (6) Report the boundary reached, sample size, estimated disparity at stopping, metric-specific assumptions, multiplicity correction, and any inconclusive outcomes.

This protocol is intentionally pre-registered: the auditor should not choose the metric, direction, or threshold after inspecting the data. Pre-registration should also state whether budget exhaustion is reported as inconclusive or converted into a terminal decision by a pre-specified truncation rule.

4.5 Operating characteristics and calibration

Simple-hypothesis error control. Let $\Lambda_t = \exp(S_t)$ denote the likelihood-ratio process for the pre-specified simple hypotheses P_0 and P_1 , and let

$$\tau = \inf\{t : \Lambda_t \geq A \text{ or } \Lambda_t \leq B\}. \quad (10)$$

Under P_0 , $(\Lambda_t)_{t \geq 0}$ is a nonnegative martingale with $\mathbb{E}_0[\Lambda_t] = 1$. Hence, by Ville's inequality,

$$\mathbb{P}_0\left(\sup_{t \geq 0} \Lambda_t \geq A\right) \leq \frac{1}{A}. \quad (11)$$

Similarly, under P_1 , $(\Lambda_t^{-1})_{t \geq 0}$ is a nonnegative martingale and

$$\mathbb{P}_1\left(\inf_{t \geq 0} \Lambda_t \leq B\right) = \mathbb{P}_1\left(\sup_{t \geq 0} \Lambda_t^{-1} \geq B^{-1}\right) \leq B. \quad (12)$$

These inequalities imply the conservative anytime bounds

$$\mathbb{P}_0(\tau_A < \infty) \leq A^{-1}, \quad \mathbb{P}_1(\tau_B < \infty) \leq B, \quad (13)$$

where τ_A and τ_B denote upper- and lower-boundary crossings, respectively. Thus, exact anytime-valid level- α control of upper crossings can be obtained by setting $A = 1/\alpha$. Wald's classical choices $A = (1 - \beta)/\alpha$ and $B = \beta/(1 - \alpha)$ instead target Type I and Type II risks jointly and are accurate when boundary overshoot is small. This distinction is important for fairness audits: the martingale argument explains why pre-specified sequential monitoring avoids ordinary optional-stopping bias, while the nominal Wald values should still be validated under the actual composite, batched, and finite-budget audit design.

Expected stopping-time intuition. The sample efficiency of the SPRT is governed by the expected log-likelihood increment. This follows from the additivity of the log-likelihood ratio: under a fixed data-generating distribution, each relevant observation contributes an expected increment equal to the Kullback-Leibler divergence between the corresponding design distributions. The expected time to hit a boundary is therefore approximately the boundary height divided by this drift, with the numerator adjusted for the probabilities of crossing the upper and lower boundaries. Under the alternative design point,

$$\mathbb{E}_1[S_t - S_{t-1}] = D_{\text{KL}}(P_1 \| P_0), \quad (14)$$

whereas under the null design point,

$$\mathbb{E}_0[S_t - S_{t-1}] = -D_{\text{KL}}(P_0 \| P_1). \quad (15)$$

Ignoring boundary overshoot, the approximate expected stopping times are

$$\mathbb{E}_1[\tau] \approx \frac{(1 - \beta) \log A + \beta \log B}{D_{\text{KL}}(P_1 \| P_0)}, \quad \mathbb{E}_0[\tau] \approx \frac{(1 - \alpha) \log B + \alpha \log A}{-D_{\text{KL}}(P_0 \| P_1)}. \quad (16)$$

Consequently, audits stop quickly when the true behavioral distribution is well separated from the design null, and require more samples near the indifference region. This explains why smaller materiality thresholds increase the sample sizes observed in Section 5. For two-group audits, $D_{\text{KL}}(P_1 \| P_0)$ should be interpreted as the expected log-likelihood increment under the actual sampling design, including the group-sampling probabilities and, for conditional metrics, the probability that a record is metric-relevant.

Finite-budget calibration. With a maximum audit budget T_{max} , the operating characteristics should be defined for the actual terminal rule. If $d_{\text{trunc}} \in \{0, 1\}$ denotes a pre-specified decision when the budget is exhausted, where 1 means flagging a violation and 0 means no violation, then

$$\alpha_{\text{fin}} = \mathbb{P}_0\{\tau \leq T_{\text{max}}, S_{\tau} \geq \log A\} + \mathbb{P}_0\{\tau > T_{\text{max}}, d_{\text{trunc}} = 1\}, \quad (17)$$

while

$$\beta_{\text{fin}} = \mathbb{P}_1\{\tau \leq T_{\text{max}}, S_{\tau} \leq \log B\} + \mathbb{P}_1\{\tau > T_{\text{max}}, d_{\text{trunc}} = 0\}. \quad (18)$$

The notation α_{fin} and β_{fin} emphasizes that these are realized operating characteristics under the finite-budget protocol, not necessarily the nominal inputs α and β . If no terminal decision is made at T_{max} , the exhaustion event should instead be reported as inconclusive. This distinction is crucial for fairness audits: data exhaustion is neither a violation finding nor a fairness certificate. Empirical calibration can be performed by simulating the complete planned audit before deployment. The auditor specifies grids of null parameter values Θ_0 and alternative values Θ_1 , including plausible group prevalences, sampling-without-replacement effects, batch sizes, and truncation rules. For each grid point, the full sequential procedure is repeated many times with the same boundaries and multiplicity correction planned for deployment. The calibrated Type I error is the largest simulated flag rate over Θ_0 , and the calibrated Type II error is the largest simulated lower-boundary or truncation rate over Θ_1 . If these risks exceed the targets, the boundaries, batch size, or error allocation should be adjusted before the audit is run.

5 Controlled Synthetic Experiments

5.1 Synthetic black-box model

We first construct a controlled model whose bias is known. The population contains two groups with equal suitability rates, $\mathbb{P}(Y = 1 | A = 1) = \mathbb{P}(Y = 1 | A = 2) = 0.8$, but the classifier applies group-dependent selection probabilities:

$$\begin{aligned} \mathbb{P}(\hat{Y} = 1 | A = 1, Y = 1) &= 0.8, & \mathbb{P}(\hat{Y} = 1 | A = 1, Y = 0) &= 0.25, \\ \mathbb{P}(\hat{Y} = 1 | A = 2, Y = 1) &= 0.5, & \mathbb{P}(\hat{Y} = 1 | A = 2, Y = 0) &= 0.10. \end{aligned}$$

Thus, both groups have the same underlying suitability distribution, but group 1 receives favorable predictions more often. We generate 1,000 individuals per group. Table 2 reports the realized outcome counts, and Figure 1 visualizes the group-level imbalance.

5.2 Fixed-sample reference

For comparison, we compute a conventional fixed-sample reference based on the normal approximation to the two-proportion test:

$$n = \frac{(Z_{1-\beta} + Z_{1-\alpha/2})^2 [p_a(1-p_a) + p_b(1-p_b)]}{(p_b - p_a)^2}. \quad (19)$$

Table 2. Outcome distribution of the synthetic black-box model.

Group / suitability	Positive prediction	Negative prediction
Group 1, suitable	639	164
Group 1, unsuitable	62	135
Group 2, suitable	388	396
Group 2, unsuitable	24	192
Group 1 total	701	299
Group 2 total	412	588

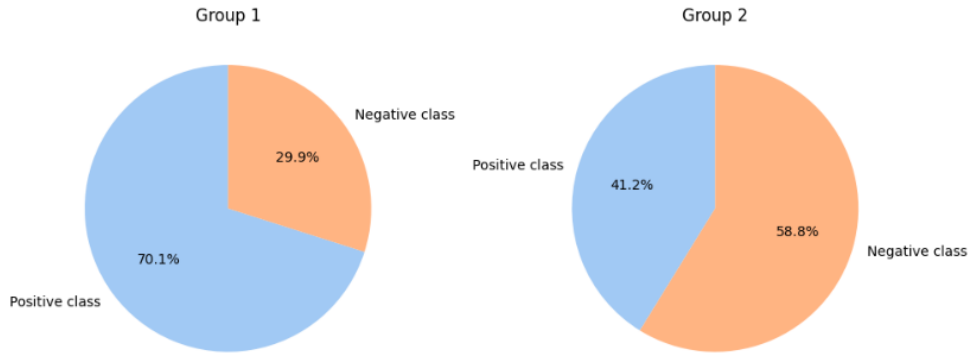


Fig. 1. Positive and negative predictions by group in the synthetic model. Although suitability rates are equal by construction, the classifier favors group 1.

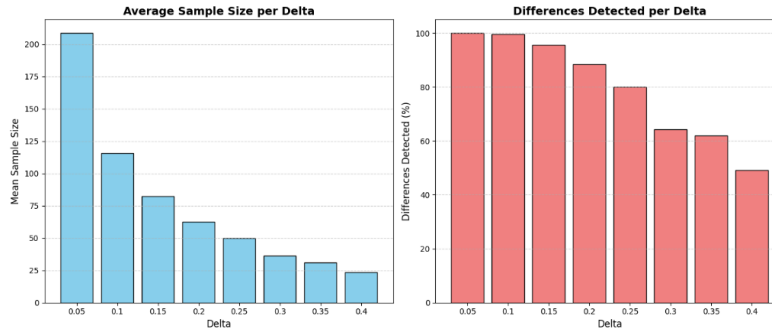
Equation (19) gives a per-group fixed-sample requirement under the anticipated probabilities. In the synthetic setting, these probabilities are known, so this is an optimistic benchmark. After accounting for the two groups, the corresponding total reference sample sizes are 100 observations for statistical parity, 72 for equal opportunity, and, for equalized odds, 88 for the TPR component and 236 for the FPR component. Since the FPR component is the limiting one, we use 236 observations as the reference sample size for equalized odds. For conditional metrics, the total number of raw audit records may be larger because only records satisfying the conditioning event contribute to the relevant Bernoulli contrast. When reporting sample savings, audit reports should distinguish among per-arm requirements, total records consumed, and metric-relevant records: statistical parity uses every audit record, whereas equal opportunity and equalized odds use only records in the relevant conditional populations. The dashed references in Figure 3 follow the implementation benchmark used in the experiments; in deployment, auditors should state explicitly whether comparisons are made against per-arm, total-record, or conditionally relevant fixed-sample budgets.

5.3 Effect of the tolerance threshold

The tolerance threshold δ controls the practical meaning of the audit. Table 3 reports 1,000 statistical-parity SPRT simulations for different values of δ .

Table 3. Avg. sample and detections for different δ values under statistical parity, based on 1,000 simulations.

δ	Avg. sample	Detections
0.05	208.57	1000
0.10	115.55	996
0.15	82.32	956
0.20	62.44	884
0.25	50.03	800
0.30	36.16	642
0.35	31.23	620
0.40	23.52	491

Fig. 2. Effect of the tolerance threshold δ on average sample size and detection rate. Smaller thresholds detect subtler disparities but require more observations.

The qualitative effect of δ is straightforward: smaller thresholds increase sensitivity but require larger expected sample sizes, whereas larger thresholds shorten audits but may miss practically relevant disparities. Thus, δ should be treated as a normative and institutional parameter rather than as a purely statistical tuning constant.

5.4 Sequential results across metrics

We next fix $\delta = 0.2$, $\alpha = 0.05$, and $1 - \beta = 0.8$, and run 10,000 simulations for each metric. The summary in Table 4 reports the number of runs in which the unfair model was detected and the average number of consumed audit records. Figure 3 shows the corresponding sample-size distributions.

Monte Carlo uncertainty is small for the main detection-rate comparisons. For example, the statistical-parity detection rate is 89.47% with an approximate 95% Monte Carlo interval of [88.87%, 90.07%], and the equalized-odds detection rate is 99.63% with interval [99.51%, 99.75%].

The sequential audits usually stop well before the fixed-sample reference. Statistical parity is the most sample-efficient metric in this design because all observations are informative. Equal opportunity discards observations with $Y = 0$, so some samples do not update the likelihood ratio. Equalized odds requires testing two error-rate components; it is therefore more stringent and consumes more samples on average, but it detects the synthetic violation most reliably because the unfair model differs in both TPR and FPR.

Table 4. Synthetic audit results across fairness metrics with $\delta = 0.2$. Each row reports 10,000 sequential simulations. “Below ref.” counts runs that consumed fewer observations than the fixed-sample reference under the implementation benchmark described in Section 5. The column uses the same reference line shown in Figure 3; it is an implementation comparison rather than a universal efficiency guarantee.

Metric	Detections	Avg. sample	No-flag avg.	Below ref.	Detected below ref.
Statistical parity	8,947	63.53	18.48	8,504	7,461
Equal opportunity	8,721	68.01	18.89	6,282	5,038
Equalized odds	9,963	98.15	39.35	9,877	9,840

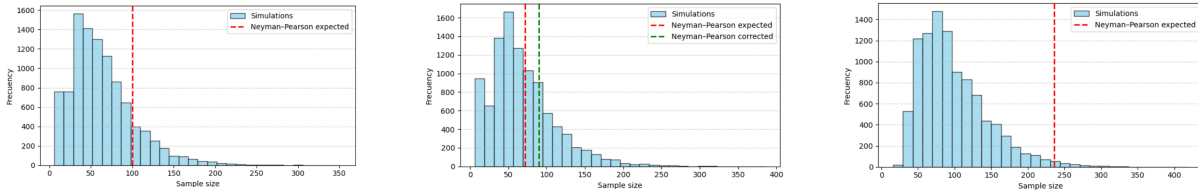


Fig. 3. Sample-size distributions for statistical parity, equal opportunity, and equalized odds. The dashed references correspond to the fixed-sample Neyman-Pearson benchmarks used in the implementation.

The simulations also reveal a practical risk. Some missed detections stop very early, when initial random fluctuations temporarily favor the null. This is an expected cost of any sequential procedure that is allowed to stop early. In practice, an auditor can mitigate this by imposing a minimum initial batch, calibrating the stopping boundaries under the actual sampling protocol, or using a truncated SPRT when the audit budget is finite [18].

5.5 Fair-model stress test for equalized odds

The equalized-odds experiment is deliberately demanding because it tests two conditional error-rate components. To check whether the apparent detection strength in the unfair case is accompanied by calibration problems under the null, we ran an additional experiment with a fair synthetic classifier whose relevant group-conditional event probabilities are equal by construction.

This stress test is important for the main methodological claim. The false-flag rate is approximately 9.4%, above the nominal 5% target, with an approximate 95% Monte Carlo interval of [7.6%, 11.2%]. This is consistent with family-wise error accumulation: two component tests run at level 0.05 can yield a family-wise false-positive probability near $1 - (1 - 0.05)^2 = 9.75\%$.

Table 5. Equalized-odds audit on a fair synthetic classifier.

Outcome	Count / value
Lower-boundary no-violation decisions	906
False bias flags	94
False-flag rate	9.4%
Average sample size	203.09
Average sample, no-violation decisions	141.91
Average sample, false flags	209.44

6 German Credit Case Study

We next evaluate the protocol on the German Credit dataset, which contains 1,000 loan applicants with a binary creditworthiness label and demographic attributes including gender [11]. The dataset is imbalanced with respect to gender, with 310 female and 690 male applicants. We train logistic-regression classifiers on 500 examples and reserve 500 examples for auditing. Credit scoring is a high-stakes allocation setting and is explicitly relevant to European AI governance debates, so we use a stricter threshold $\delta = 0.1$ for the statistical-parity audit.

We compare three models. The baseline uses all available features, including gender. The second model removes the sensitive attribute from the feature set. The third model uses Fairlearn to train under a demographic-parity constraint [21]. All three systems use the same train/audit split. Numerical variables are standardized and categorical variables are encoded before fitting logistic regression; the fairness-aware model is trained with a demographic-parity constraint, and the sequential audit samples from the held-out audit set without using audit outcomes to refit the model or update p_0 . This separation is important because using sequential audit data to tune the design point or mitigation method would invalidate the reported stopping behavior. Because the German Credit audit samples from a finite held-out set, sampling without replacement induces finite-population dependence. We therefore interpret the Bernoulli SPRT as an operational audit approximation and rely on simulation under the realized sampling protocol to assess its operating characteristics. In a deployment setting with streaming or independently sampled audit records, the i.i.d. Bernoulli approximation is more direct. Each audit instantiates Eq. (4) as

$$H_0 : |p_{\text{male}} - p_{\text{female}}| \leq \delta_0 \quad \text{versus} \quad H_1 : |p_{\text{male}} - p_{\text{female}}| \geq \delta_1, \quad (20)$$

with $\delta_0 = 0$ and $\delta_1 = 0.1$, where $p_g = \Pr(\hat{Y} = 1 | A = g)$. We run 10,000 simulations for each configuration. Table 6 reports bias flags, lower-boundary no-flag decisions, and data-exhaustion outcomes. These outcomes partition the Monte Carlo runs:

$$N_{\text{flag}} + N_{\text{no flag}} + N_{\text{insuff}} = N_{\text{sim}}.$$

Thus, $N_{\text{no flag}}$ is computed by subtraction. A no-flag outcome means that the lower boundary was reached; it is not a proof of fairness beyond the pre-specified audit design.

The baseline model is frequently flagged, indicating a substantial statistical-parity disparity. Removing gender from the feature set does not eliminate the problem: the flag rates remain high, consistent with the presence of proxy variables correlated with gender. This supports a common lesson from fairness interventions: excluding the protected attribute is not the same as removing information about group membership.

Table 6. German Credit sequential audits. “Bias flags” is the number of simulations, out of $N_{\text{sim}} = 10,000$, in which the audit flagged a fairness violation. “No flag” counts lower-boundary no-violation decisions and is computed as $N_{\text{sim}} - N_{\text{flag}} - N_{\text{insuff}}$. “Insuff.” counts runs that exhausted the audit budget before a boundary was crossed.

Model	(α, β) , batch	Avg. sample	Bias flags	No flag	Insuff.	Flag rate
Baseline	(0.05,0.20), 15	243.46	7,718	2,281	1	77.18%
Baseline	(0.05,0.10), 15	286.72	8,835	1,164	1	88.35%
Baseline	(0.10,0.20), 15	181.29	7,543	2,457	0	75.43%
Baseline	(0.10,0.10), 15	219.89	8,660	1,340	0	86.60%
Baseline	(0.05,0.20), 5	209.87	6,962	3,038	0	69.62%
Baseline	(0.05,0.10), 5	255.75	8,249	1,751	0	82.49%
Baseline	(0.10,0.20), 5	155.85	6,943	3,057	0	69.43%
Baseline	(0.10,0.10), 5	193.89	8,128	1,872	0	81.28%
No sensitive attr.	(0.05,0.20), 15	256.81	7,580	2,383	37	75.80%
No sensitive attr.	(0.05,0.10), 15	303.78	8,452	1,285	263	84.52%
No sensitive attr.	(0.10,0.20), 15	190.82	7,404	2,596	0	74.04%
No sensitive attr.	(0.10,0.10), 15	233.92	8,673	1,327	0	86.73%
No sensitive attr.	(0.05,0.20), 5	220.77	6,858	3,119	23	68.58%
No sensitive attr.	(0.05,0.10), 5	272.01	8,010	1,846	144	80.10%
No sensitive attr.	(0.10,0.20), 5	163.48	6,757	3,243	0	67.57%
No sensitive attr.	(0.10,0.10), 5	208.15	8,163	1,837	0	81.63%
Fairness-aware	(0.05,0.20), 15	215.28	276	9,195	529	2.76%
Fairness-aware	(0.05,0.10), 15	300.76	232	6,600	3,168	2.32%
Fairness-aware	(0.10,0.20), 15	191.42	934	8,785	281	9.34%
Fairness-aware	(0.10,0.10), 15	274.03	901	6,529	2,570	9.01%
Fairness-aware	(0.05,0.20), 5	180.82	317	9,365	318	3.17%
Fairness-aware	(0.05,0.10), 5	271.10	321	7,102	2,577	3.21%
Fairness-aware	(0.10,0.20), 5	157.29	1,015	8,816	169	10.15%
Fairness-aware	(0.10,0.10), 5	241.37	994	7,089	1,917	9.94%

The fairness-aware model is flagged substantially less often, especially for $\alpha = 0.05$, where the flag rate is approximately 2–3%. At the same time, many runs become inconclusive when the audit requires high power ($\beta = 0.10$) under a small threshold and a limited audit set. This illustrates a useful distinction for reporting: a low flag rate is not identical to a formal proof of fairness when many audits terminate without a decision. The added no-flag column separates these lower-boundary decisions from data-exhaustion outcomes, which prevents inconclusive audits from being interpreted as fairness certificates. In a deployment setting, such outcomes should be reported as inconclusive or handled by a truncated sequential rule rather than silently converted into acceptances.

Table 7 reports predictive performance. The fairness-aware model reduces statistical-parity flags, but its accuracy and F1 score are slightly lower than those of the baseline. The drop is small in this experiment, but the example illustrates why audit reports should present both predictive and fairness-relevant evidence.

Finally, the case study reinforces that fairness mitigation is metric-specific. The Fairlearn model was trained to reduce demographic-parity violations, so the statistical-parity audit is aligned with the training objective. When the audit metric changes, the same model can still fail under other criteria. This is not a contradiction; it is a reminder that fairness claims should be stated relative to a specified metric and context.

Table 7. Predictive performance of the audited German Credit models.

Model	Accuracy	Precision	Recall	F1 score
Baseline	0.77	0.80	0.88	0.84
No sensitive attribute	0.76	0.80	0.88	0.84
Fairness-aware	0.75	0.79	0.87	0.83

7 Discussion and Limitations

What the audit can and cannot certify. A sequential black-box audit can provide statistically disciplined evidence about a specified behavioral property. It cannot establish that a system is fair in all respects, nor can it identify causal mechanisms without additional access to data, model internals, or deployment processes. We therefore view the proposed method as an efficient screening and monitoring tool within a broader accountability process.

Composite hypotheses and calibration. Classical SPRT guarantees apply to simple hypotheses. Fairness audits typically involve composite hypotheses such as $|D(f)| \leq \delta_0$ versus $|D(f)| \geq \delta_1$. The practical implementation must therefore choose design points, use a generalized or mixture likelihood ratio, or calibrate the test by simulation. A natural composite alternative is the generalized likelihood-ratio statistic

$$S_t^{\text{GLR}} = \log \frac{\sup_{(p_a, p_b): |p_a - p_b| \geq \delta_1} L_t(p_a, p_b)}{\sup_{(p_a, p_b): |p_a - p_b| \leq \delta_0} L_t(p_a, p_b)}. \quad (21)$$

where $L_t(p_a, p_b)$ is the Bernoulli likelihood after t relevant observations. This statistic avoids committing to a single design point p_0 , but its finite-sample stopping boundaries generally require calibration or time-uniform concentration tools. Our experiments use simple design points for transparency. This makes the procedure easy to audit, but it also means that the nominal Wald boundaries should be understood as design targets rather than automatic finite-sample compliance guarantees for every composite fairness problem. For a composite fairness audit, calibration should be performed over the sets

$$\Theta_0 = \{(p_a, p_b) : |p_a - p_b| \leq \delta_0\}, \quad \Theta_1 = \{(p_a, p_b) : |p_a - p_b| \geq \delta_1\}. \quad (22)$$

The relevant finite-budget risks are therefore

$$\alpha_{\max} = \sup_{\theta \in \Theta_0} \mathbb{P}_{\theta} \{\text{flag a violation before } T_{\max}\}, \quad (23)$$

and

$$\beta_{\max} = \sup_{\theta \in \Theta_1} \mathbb{P}_{\theta} \{\text{return no violation before } T_{\max} \text{ or under the truncation rule}\}. \quad (24)$$

This worst-case view is more appropriate for compliance-oriented audits than calibration at a single design point, because the auditor rarely knows the true common event probability in advance. Before deployment, the achieved Type I and Type II error rates should be checked under the intended sampling design, especially when using batches or testing multiple components such as equalized odds. This check should use the same sample-counting convention that will be reported in the audit: per-arm counts, total records consumed, and metric-relevant records can lead to different apparent efficiency comparisons.

Normative role of δ . The threshold δ determines which disparities the audit treats as material. This choice should not be optimized only for statistical convenience. In credit, hiring, policing, healthcare, and public-service allocation, small disparities may have large cumulative effects. In lower-stakes or exploratory contexts, a larger threshold may be appropriate. In practice, auditors should justify δ using the deployment domain, legal or institutional materiality standards, historical disparity levels, the size of the affected population, and the relative costs of false violation flags and missed disparities. The paper’s main methodological recommendation is to make this threshold explicit and contestable.

Finite data and inconclusive decisions. Sequential tests are attractive because they can stop early, but finite audit budgets create a third outcome: inconclusive. This outcome is substantively important. Inconclusive audits should trigger additional data collection, a pre-specified truncated sequential decision, or a narrower audit claim. They should not be reported as evidence that the system is fair.

Minimum reporting checklist. A reproducible audit report should disclose: the audited population and sampling frame; protected groups; fairness metric and justification; materiality threshold δ ; α, β , batch size, maximum budget, design probabilities such as p_0 , Monte Carlo intervals, and any minimum-batch rule; multiplicity correction; decision, stopping time, estimated disparity, and inconclusive outcomes; and whether operating characteristics were calibrated under the realized sampling protocol.

8 Conclusion

We presented a sequential hypothesis-testing framework for black-box fairness auditing. By expressing group fairness checks as pre-registered Bernoulli contrasts and applying the SPRT, the audit can often reach decisions with fewer observations than fixed-sample tests while retaining an anytime-valid likelihood-ratio justification under simple-hypothesis design conditions, with finite-sample risks calibrated for composite and batched audits. Synthetic experiments show sample-efficiency gains across statistical parity, equal opportunity, and equalized odds. A German Credit case study shows that the framework detects disparities in a baseline model, that feature removal alone is insufficient, and that fairness-aware training reduces statistical-parity flags while introducing a modest performance trade-off. The broader lesson is that statistically rigorous auditing requires more than a test statistic: it requires a justified metric, a transparent materiality threshold, calibrated error rates, and careful reporting of inconclusive outcomes. Future work should develop multiplicity-aware composite sequential tests, extend the framework to intersectional and multi-class settings, and integrate black-box audits with causal and organizational audit procedures. The same sequential structure may also apply to other black-box behavioral requirements expressible as Bernoulli or bounded-loss contrasts, including safety violations, content-policy failures, and reliability regressions.

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References

- [1] Solon Barocas, Moritz Hardt, and Arvind Narayanan. 2023. *Fairness and Machine Learning: Limitations and Opportunities*. MIT Press.
- [2] Yoav Benjamini and Yosef Hochberg. 1995. Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. *Journal of the Royal Statistical Society Series B* 57, 1 (1995), 289–300. doi:10.1111/j.2517-6161.1995.tb02031.x
- [3] Yoav Benjamini and Daniel Yekutieli. 2001. The Control of the False Discovery Rate in Multiple Testing under Dependency. *The Annals of Statistics* 29, 4 (2001), 1165–1188. doi:10.1214/aos/1013699998
- [4] Emily Black, Manish Raghavan, and Solon Barocas. 2022. Model Multiplicity: Opportunities, Concerns, and Solutions. In *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency*. ACM, 850–863. doi:10.1145/3531146.3533149
- [5] Stephen Casper, Carson Ezell, Charlotte Siegmann, Noam Kolt, Taylor Lynn Curtis, Andreas Haupt, Kevin Wei, Jérémy Scheurer, Marius Hobbhahn, Lee Sharkey, et al. 2024. Black-Box Access is Insufficient for Rigorous AI Audits. In *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency*. ACM, 2254–2272. doi:10.1145/3630106.3659037
- [6] Sarah H. Cen and Rohan Alur. 2024. From Transparency to Accountability and Back: A Discussion of Access and Evidence in AI Auditing. In *Proceedings of the 2024 ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization*. ACM. doi:10.1145/3689904.3694711
- [7] Ben Chugg, Santiago Cortes-Gomez, Bryan Wilder, and Aaditya Ramdas. 2023. Auditing Fairness by Betting. *arXiv preprint arXiv:2305.17570* (2023). doi:10.48550/arXiv.2305.17570
- [8] European Parliament and Council of the European Union. 2024. Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 Laying Down Harmonised Rules on Artificial Intelligence and Amending Certain Union Legislative Acts (Artificial Intelligence Act). Official Journal of the European Union L, 12 July 2024. <https://eur-lex.europa.eu/eli/reg/2024/1689/oj>
- [9] Antonio Ferrara, Francesco Cozzi, Alan Perotti, André Panisson, and Francesco Bonchi. 2025. Size-Adaptive Hypothesis Testing for Fairness. *arXiv preprint arXiv:2506.10586* (2025). doi:10.48550/arXiv.2506.10586
- [10] Moritz Hardt, Eric Price, and Nati Srebro. 2016. Equality of Opportunity in Supervised Learning. In *Advances in Neural Information Processing Systems*, Vol. 29. 3315–3323.
- [11] Hans Hofmann. 1994. Statlog (German Credit Data). UCI Machine Learning Repository. doi:10.24432/C5NC77
- [12] Steven R. Howard, Aaditya Ramdas, Jon McAuliffe, and Jasjeet Sekhon. 2020. Time-Uniform Chernoff Bounds via Nonnegative Supermartingales. *Probability Surveys* 17 (2020), 257–317. doi:10.1214/18-PS321
- [13] Ramesh Johari, Pete Koomen, Leonid Pekelis, and David Walsh. 2022. Always Valid Inference: Continuous Monitoring of A/B Tests. *Operations Research* 70, 3 (2022), 1806–1821. doi:10.1287/opre.2021.2135
- [14] Jon Kleinberg, Jens Ludwig, Sendhil Mullainathan, and Cass R. Sunstein. 2018. Discrimination in the Age of Algorithms. *Journal of Legal Analysis* 10, 1 (2018), 113–174. doi:10.1093/jla/laz001
- [15] Jon Kleinberg, Sendhil Mullainathan, and Manish Raghavan. 2017. Inherent Trade-Offs in the Fair Determination of Risk Scores. In *Proceedings of the 8th Innovations in Theoretical Computer Science Conference*. Schloss Dagstuhl – Leibniz-Zentrum für Informatik. doi:10.4230/LIPIcs.ITCS.2017.43
- [16] Jakob Mökander. 2023. Auditing of AI: Legal, Ethical and Technical Approaches. *Digital Society* 2, 3 (2023). doi:10.1007/s44206-023-00074-y
- [17] Inioluwa Deborah Raji, Andrew Smart, Rebecca N. White, Margaret Mitchell, Timnit Gebru, Ben Hutchinson, Jamila Smith-Loud, Daniel Theron, and Parker Barnes. 2020. Closing the AI Accountability Gap: Defining an End-to-End Framework for Internal Algorithmic Auditing. In *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*. ACM, 33–44. doi:10.1145/3351095.3372873
- [18] Sawasd Tantaratana and John B. Thomas. 1977. Truncated Sequential Probability Ratio Test. *Information Sciences* 13, 3 (1977), 283–300. doi:10.1016/0020-0255(77)90050-0
- [19] Bahar Taskesen, Jose Blanchet, Daniel Kuhn, and Viet Anh Nguyen. 2021. A Statistical Test for Probabilistic Fairness. In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*. ACM, 648–665. doi:10.1145/3442188.3445927
- [20] Abraham Wald. 1947. *Sequential Analysis*. John Wiley & Sons, New York, NY.
- [21] Hilde Weerts, Miroslav Dudík, Richard Edgar, Adrin Jalali, Roman Lutz, and Michael Madaio. 2023. Fairlearn: Assessing and Improving Fairness of AI Systems. *Journal of Machine Learning Research* 24, 257 (2023), 1–8.